Chapter test

25 Body defence mechanisms

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following is ***not*** a physical barrier in the human body?

   A. epidermis of the skin
   B. tears produced by tear glands
   C. mucus produced in the trachea
   D. hair in the nostrils ☐

2. Peter had a cut on his finger. He found that the skin surrounding the wound had become red and swollen. Which of the following is a possible reason for this?

   A. The skin turned red because the body temperature increased.
   B. The skin turned red because there was infection.
   C. The skin became swollen because phagocytes were producing antibodies.
   D. The skin became swollen because there was an accumulation of tissue fluid. ☐

3. Phagocytes kill pathogens in two main steps. Which of the following correctly describes the steps?

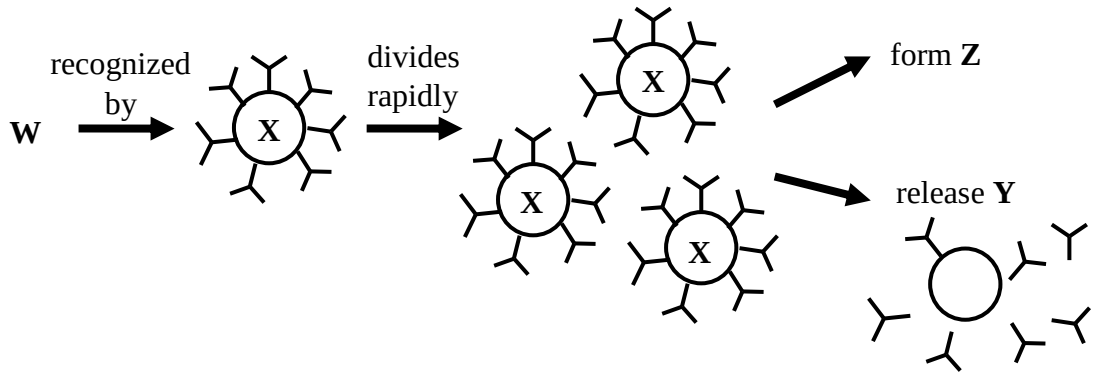
	Step 1	***Step 2***	
A.	ingest the pathogens by endocytosis	digest the pathogens with antibodies	
B.	ingest the pathogens by endocytosis	digest the pathogens with lysozyme	
C.	ingest the pathogens by active transport	digest the pathogens with antibodies	
D.	ingest the pathogens by active transport	digest the pathogens with lysozyme	☐

4. Which of the following descriptions about antibodies is ***incorrect***?

   A. They can kill pathogens by carrying out phagocytosis.
   B. They are produced by B cells.
   C. They are protein in nature.
   D. The specificity of an antibody is due to its antigen-binding site. ☐

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★ 5. The diagram below shows a simplified immune response:



What do W, X, Y and Z represent respectively?

	W	***X***	***Y***	***Z***
A.	antigen	T cell	antibody	phagocyte
B.	antigen	B cell	antibody	memory cell
C.	antibody	T cell	antigen	phagocyte
D.	antibody	B cell	antigen	memory cell

6. Which type of cells is responsible for destroying viral infected cells?

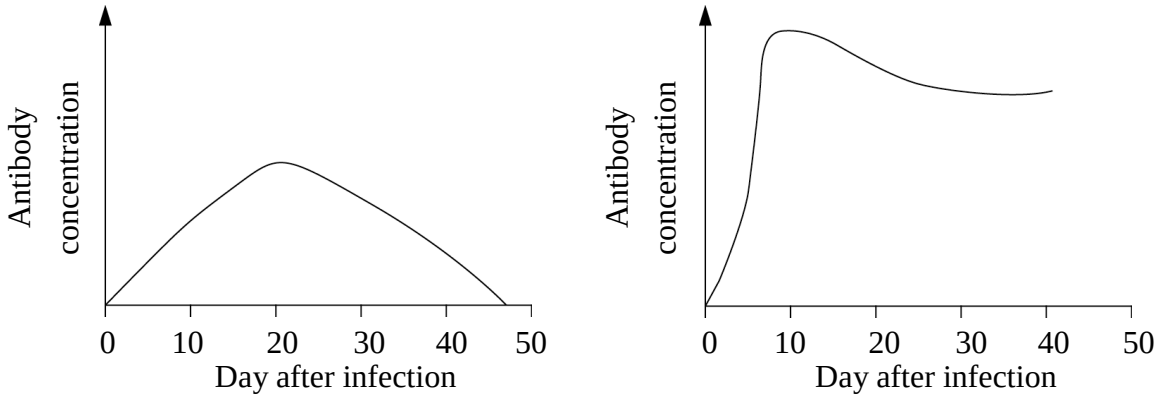
A. B cells
B. plasma cells
C. killer T cells
D. helper T cells

7. The immunity passed from a mother to a baby through breast feeding can be described as

A. active immunity acquired naturally.
B. active immunity acquired artificially.
C. passive immunity acquired naturally.
D. passive immunity acquired artificially.

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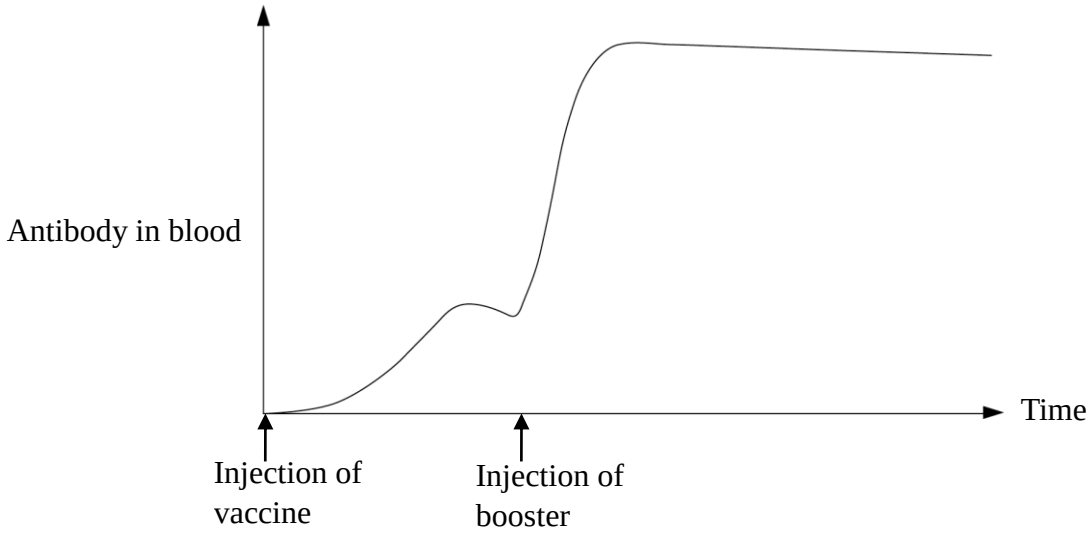
★ 8. The graphs below show the primary and secondary responses of a child when she was infected by the same virus twice.



Which of the following accounts for the difference between the two responses?

A. There are some antigens remain after the primary response.
B. There are some antibodies remain after the primary response.
C. The virus becomes weaker.
D. The virus is memorized by memory cells. □

Directions: Questions 9 and 10 refer to the graph below, which shows the changes in the antibody level after injections of a vaccine and the booster.



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★ 9. The vaccine injected may contain

(1) memory cells
(2) antigens
(3) antibodies

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 10. Why did the antibody level rise drastically after the injection of the booster?

A. The booster stimulates rapid antibody replication in the human body.
B. The antibody is specific to the substance in the booster.
C. Memory cells specific to the substance in the booster are already present in the human body.
D. The substance in the booster is rapidly neutralized by the antibodies. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	25.1	25.1	25.1	25.2	25.2	25.2	25.2	25.2	25.2	25.2

B. Problem solving (17 marks)

11. Bleeding occurs when the skin is cut. Shortly after, the blood clots and bleeding stops.

(a) Explain why bleeding is important to body defence as long as it stops soon. (1 mark)

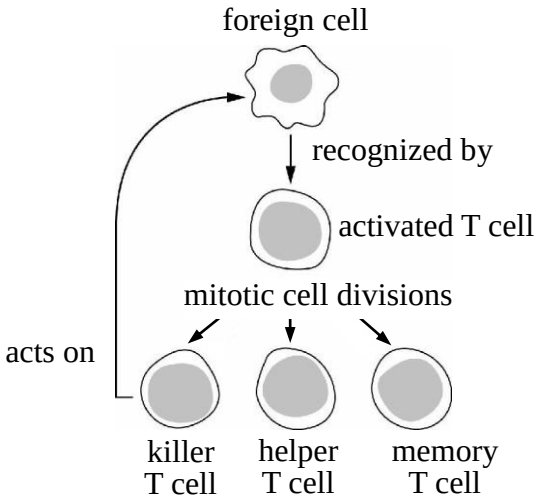
(b) Outline the process of blood clotting. (4 marks)

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(c) Bacteria may enter our body before the wound is sealed. Briefly describe the role of white blood cells in killing the invading bacteria. (2 marks)

(Total: 7 marks)

★ 12. The diagram below shows how a foreign cell elicits a specific immune response in the human body.



(a) List the ***two*** characteristics of specific immune responses. (2 marks)

(b) Name the molecules on the surface of a foreign cell that stimulate T cells. How do T cells recognize these molecules? (3 marks)

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(c) What are the functions of helper T cells in specific immune responses? (2 marks)

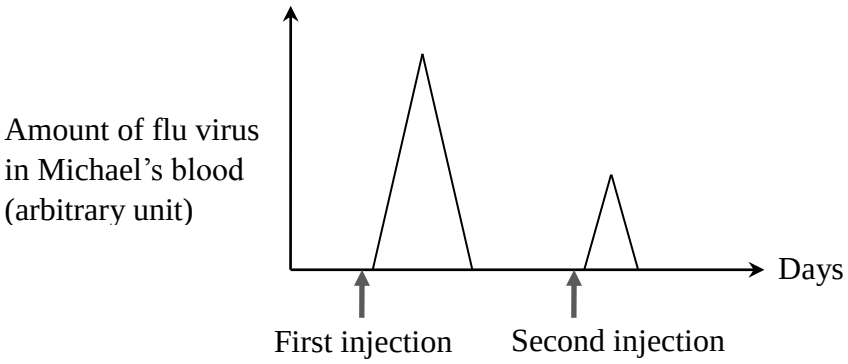
(d) What is the action of killer T cells? (1 mark)

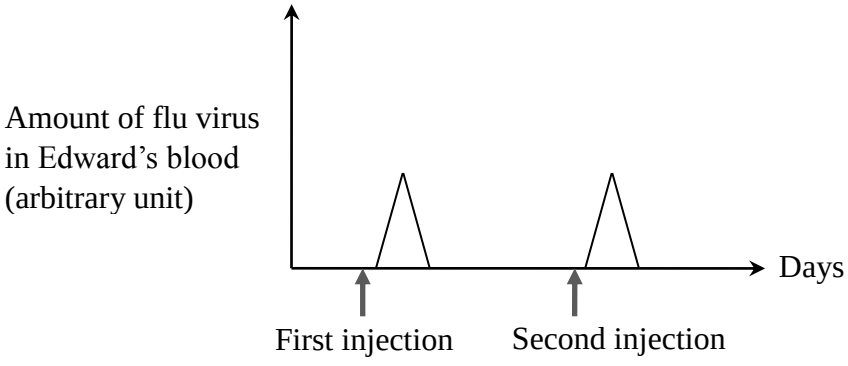
(e) Describe the function and significance of memory T cells in body defence. (2 marks)

(Total: 10 marks)

C. Exam challenge (13 marks)

★★ 13. Michael and Edward are two similar teenagers with normal immune system. In a medical test, Michael and Edward were injected with the same amount of flu virus twice within two months. The amount of flu virus in their blood was measured after each injection. The results are shown in the graph below.





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(a) Describe and explain the changes in the amount of flu virus in Michael's body after the first injection. (4 marks)

(b) Explain why the amount of flu virus in Michael's body after the second injection is different from after the first injection. (4 marks)

(c) Suggest why the amount of flu virus in Edward's body after the first injection is more or less the same as that after the second injection. (2 marks)

(d) What are the differences between active immunity and passive immunity? (3 marks)

(Total: 13 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	B	2.	D	3.	B	4.	A	5.	B
6.	C	7.	C	8.	D	9.	A	10.	C

(1 mark each)

B. Problem solving (17 marks)

11. (a) Bleeding can rush the pathogens and other dirty materials out of the wound / body. (1)

(b) When a blood vessel is damaged, platelets are attracted to the wound. (1)
Platelets and the injured tissues release signal molecules into the blood. (1)
Soluble fibrinogens are converted to insoluble fibrins. (1)
Fibrins form a network, trapping the pathogens, red blood cells, platelets and white blood cells. (1)

(c) Phagocytes engulf the bacteria by phagocytosis. (1)
B cells produce antigen-specific antibodies to destroy the bacteria. (1)

(Total: 7 marks)

12. (a) Specificity (1)
Immunological memory (1)

(b) Antigens (1)
There are receptor molecules on the surfaces of T cells. (1)
When the receptor molecules bind specifically to matching antigens, the T cells become activated. (1)

(c) Helper T cells recognize antigens (1)
and then activate B cells and killer T cells to destroy pathogens or infected cells. (1)

(d) Killer T cells bind to the cells infected by pathogens or abnormal cells (e.g. cancer cells) and destroy them directly. (1)

(e) Memory T cells will multiply and differentiate into killer T cells more quickly when the same type of foreign cell enters the body again, (1)
so as to eliminate the foreign cell more rapidly before any symptoms of infection could arise. (1)

(Total: 10 marks)

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C. Exam challenge (13 marks)

13. (a) The amount of flu virus increased after the first injection (1)
because the flu virus multiplied in Michael's body. (1)
The amount of flu virus decreased afterwards (1)
because antibodies / killer T cells were produced to destroy the flu virus. (1)

(b) The amount of flu virus after the second injection is smaller than after the first injection. (1)
After the first injection, the flu virus has been memorized by memory B cells / memory T cells. (1)
When Michael was injected with the same flu virus again, memory B cells became plasma cells to produce antibodies more rapidly / memory T cells became killer T cells more rapidly. (1)
As a result, the flu virus was killed more rapidly. (1)

(c) Edward had infected flu virus before the two injections. (1)
As a result, both injections led to secondary immune response. (1)

(d) Any ***three*** of the following, one mark for each comparison: (max. 3)

Active immunity	Passive immunity
Antibodies are produced by the immune system	Ready-made antibodies are transferred into the body
Takes time to develop and thus does not provide immediate immunity	Provides immediate immunity
Lasts for a long time	Short-lived
Naturally acquired by recovering from a disease / Artificially acquired by vaccination	Naturally acquired from the mother to the foetus or infant through the placenta or by breast feeding / Artificially acquired by injection of serum

(Total: 13 marks)

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